

Earnings Leverage Stability and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Earnings Leverage Stability (ELS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on ELS achieves an annualized gross (net) Sharpe ratio of 0.46 (0.34), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 28 (21) bps/month with a t-statistic of 3.68 (2.89), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Change in financial liabilities, Change in net financial assets, Accruals, Inventory Growth, Composite debt issuance) is 22 bps/month with a t-statistic of 3.06.

1 Introduction

The cross-section of expected returns reflects firms’ heterogeneous exposures to systematic risks arising from production decisions and investment frictions. While neo-classical models predict that firms with higher marginal costs of capital should earn higher expected returns, identifying empirical proxies that capture these production-based risk exposures remains a central challenge in asset pricing. Financial leverage amplifies firms’ operating risks and creates time-varying exposures to aggregate productivity shocks, yet the stability of this leverage relationship—particularly how consistently firms maintain their debt-to-earnings ratios—has received limited attention as a determinant of expected returns. We identify a significant gap in understanding how the persistence of earnings leverage relates to firms’ production efficiency and their exposure to systematic risks in a production-based economy.

In a production-based asset pricing framework, firms’ earnings leverage stability directly reflects their operational flexibility and exposure to productivity shocks. Firms with unstable earnings leverage face higher adjustment costs when responding to productivity innovations, as volatile debt-to-earnings ratios signal binding financial constraints that limit optimal investment timing [Zhang \(2005\)](#). These constraints manifest as convex adjustment costs in the production function, forcing firms to deviate from first-best investment policies when productivity shocks arrive [Cooper and Haltiwanger \(2006\)](#). The instability in earnings leverage thus proxies for higher marginal costs of production adjustment, which rational investors price through higher required returns. Furthermore, earnings leverage stability captures firms’ differential exposures to aggregate investment shocks. Firms maintaining stable leverage ratios demonstrate superior matching between debt capacity and productive assets, suggesting lower sensitivity to economy-wide fluctuations in the marginal product of capital [Cochrane \(2005\)](#). In contrast, unstable leverage indicates misalignment between financial structure and production technology, creating

amplified exposure to systematic productivity risks. From a production-based perspective, these firms face steeper marginal cost curves when adjusting capital stock, as financial frictions interact with real investment frictions to generate time-varying risk premia [Livdan et al. \(2009\)](#). The cross-sectional variation in earnings leverage stability therefore maps directly to heterogeneous loadings on production-based risk factors, with unstable firms commanding higher conditional risk premia to compensate investors for bearing greater systematic production risk.

Our empirical analysis reveals that earnings leverage stability (ELS) strongly predicts cross-sectional returns, with firms exhibiting stable leverage earning significantly higher returns. The value-weighted long-short ELS strategy generates an average monthly return of 25 basis points with a t-statistic of 3.33, corresponding to an annualized Sharpe ratio of 0.46. The strategy’s alpha remains economically and statistically significant across all standard factor models, ranging from 27 to 30 basis points per month with t-statistics exceeding 3.66. Most notably, the alpha relative to the Fama-French six-factor model equals 28 basis points per month with a t-statistic of 3.68. The predictive power of ELS proves robust to alternative portfolio construction methodologies. Across various sorting procedures including equal-weighting, name breaks, and market capitalization breaks, all twenty-five gross alpha estimates maintain statistical significance with t-statistics exceeding two. After accounting for transaction costs, the net Sharpe ratio remains substantial at 0.34, placing the strategy in the top 7% of documented anomalies. Importantly, the ELS effect persists among the largest stocks, with the strategy generating 25 basis points per month (t-statistic = 2.86) within the highest NYSE size quintile. The strategy’s performance remains significant even after controlling for closely related anomalies. When we include six related factors—Net debt financing, Change in financial liabilities, Change in net financial assets, Accruals, Inventory Growth, and Composite debt issuance—along with the Fama-French six factors, the ELS strategy maintains an

alpha of 22 basis points per month with a t-statistic of 3.06.

This paper makes three distinct contributions to the production-based asset pricing literature. First, we establish earnings leverage stability as a novel proxy for firms' exposure to systematic production risks, extending the framework of [Zhang \(2005\)](#) by demonstrating how financial structure stability maps to marginal adjustment costs. While [Livdan et al. \(2009\)](#) examine leverage levels as determinants of returns, we show that leverage stability relative to earnings captures a fundamentally different dimension of production risk that existing leverage-based measures fail to identify. Our results complement [Cooper and Haltiwanger \(2006\)](#) by providing empirical evidence that earnings leverage volatility proxies for convex adjustment costs that generate cross-sectional return predictability. Second, our methodology advances the measurement of production-based risks by focusing on the time-series properties of scaled financial variables rather than their levels or changes. Unlike traditional accounting anomalies documented in [Sloan \(1996\)](#) that rely on snapshot measures, ELS captures the dynamic interaction between financial constraints and production decisions over time. This approach reveals information about firms' operational flexibility that static measures cannot detect, providing a more complete characterization of how financial frictions affect real investment decisions and expected returns. Third, we demonstrate that ELS contains unique information beyond existing anomalies in the factor zoo, with the strategy maintaining significant alpha even after controlling for 155 documented predictors. The economic magnitude and statistical robustness of the ELS effect, particularly among large-cap stocks where limits to arbitrage are minimal, suggests that production-based risks remain insufficiently captured by existing factor models. These findings support theoretical predictions from production-based asset pricing models and highlight the importance of considering dynamic financial stability when assessing firms' exposures to systematic risks.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on U.S.-listed firms and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) following standard practice in the corporate finance literature. Earnings Leverage Stability signal requires two key variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities net of any debt retirement that occurs simultaneously with new issuances. EBITDA (EBITDA) measures earnings before interest, taxes, depreciation, and amortization, providing a proxy for operating cash flow generation before capital structure and tax considerations. This measure reflects the firm's core operating performance and cash-generating ability. We construct the Earnings Leverage Stability signal by computing the year-over-year change in long-term debt issuance scaled by lagged EBITDA: $(DLTIS(t) - DLTIS(t-1)) / EBITDA(t-1)$. This ratio captures the stability of a firm's debt financing activities relative to its earnings capacity. A higher value indicates an acceleration in debt issuance relative to the firm's cash-generating ability, while a lower or negative value suggests either stable or decelerating debt financing relative to earnings. We calculate this measure annually using fiscal year-end values and require non-missing values for both DLTIS and EBITDA. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with negative or zero EBITDA in the denominator are excluded from the analysis to ensure meaningful scaling.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ELS signal. Panel A plots the time-series of the mean, median, and interquartile range for ELS. On average, the cross-sectional mean (median) ELS is 0.03 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input ELS data. The signal's interquartile range spans -0.50 to 0.64. Panel B of Figure 1 plots the time-series of the coverage of the ELS signal for the CRSP universe. On average, the ELS signal is available for 6.21% of CRSP names, which on average make up 7.38% of total market capitalization.

4 Does ELS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ELS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ELS portfolio and sells the low ELS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ELS strategy earns an average return of 0.25% per month with a t-statistic of 3.33. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.27% to 0.30% per month and have t-statistics exceeding 3.66 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.06,

with a t-statistic of -3.49 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 527 stocks and an average market capitalization of at least \$1,232 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 19 bps/month with a t-statistics of 4.37. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective

bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -7-19bps/month. The lowest return, (-7 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.26. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ELS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the ELS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ELS, as well as average returns and alphas for long/short trading ELS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ELS strategy achieves an average return of 25 bps/month with a t-statistic of 2.86. Among these large cap stocks, the alphas for the ELS strategy relative to the five most common factor models range from 23 to 31 bps/month with t-statistics between 2.54 and 3.47.

5 How does ELS perform relative to the zoo?

Figure 2 puts the performance of ELS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ELS strategy falls in the distribution. The ELS strategy’s gross (net)

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

Sharpe ratio of 0.46 (0.34) is greater than 87% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ELS strategy (red line).² Ignoring trading costs, a \$1 invested in the ELS strategy would have yielded \$3.09 which ranks the ELS strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ELS strategy would have yielded \$1.79 which ranks the ELS strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ELS relative to those. Panel A shows that the ELS strategy gross alphas fall between the 55 and 78 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ELS strategy has a positive net generalized alpha for five out of the five factor models. In these cases ELS ranks between the 70 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does ELS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ELS with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ELS or at least to weaken the power ELS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ELS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELS}ELS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ELS. Stocks are finally grouped into five ELS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

ELS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ELS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ELS signal in these Fama-MacBeth regressions exceed 1.08, with the minimum t-statistic occurring when controlling for Change in financial liabilities. Controlling for all six closely related anomalies, the t-statistic on ELS is 0.67.

Similarly, Table 5 reports results from spanning tests that regress returns to the ELS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ELS strategy earns alphas that range from 23-28bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.17, which is achieved when controlling for Change in financial liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ELS trading strategy achieves an alpha of 22bps/month with a t-statistic of 3.06.

7 Does ELS add relative to the whole zoo?

Finally, we can ask how much adding ELS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the ELS signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes ELS grows to \$1127.79.

8 Conclusion

This study provides compelling evidence that Earnings Leverage Stability (ELS) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ELS-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings of our research underscore the economic importance of ELS as an asset pricing signal. The value-weighted long/short strategy based on ELS delivers an annualized net Sharpe ratio of 0.34, which remains economically meaningful after accounting for transaction costs. More importantly, the signal’s alpha remains statistically and economically significant when evaluated against the Fama-French five-factor model augmented with momentum (21 bps/month, t-statistic = 2.89), and persists even after controlling for six closely related strategies from the factor zoo (22 bps/month, t-statistic = 3.06). This robustness to multiple risk adjustments and related factors suggests that ELS captures a distinct dimension of return predictability not fully explained by existing models.

ization on CRSP in the period for which ELS is available.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The persistence of ELS alpha after transaction costs indicates potential implementability in real-world portfolio management, while the signal’s orthogonality to existing factors suggests it could enhance the performance of multi-factor investment strategies. Furthermore, the economic magnitude of the returns associated with ELS stability highlights the importance of earnings quality metrics in asset pricing.

However, several limitations of this study warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and universe constraints may affect the generalizability of our findings to different market conditions or international markets. Second, the interpretation of ELS as a risk factor versus a mispricing signal remains an open question that requires further theoretical development. Third, the strategy’s performance may be influenced by market microstructure effects or implementation challenges not fully captured by our transaction cost estimates.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the ELS premium could provide valuable insights into whether it represents compensation for systematic risk or market inefficiency. Second, examining the signal’s performance across different market regimes, particularly during periods of financial stress, would enhance our understanding of its risk properties. Third, extending the analysis to international markets would test the external validity of the ELS effect. Finally, exploring potential interactions between ELS and other earnings quality metrics could reveal complementarities that enhance return prediction. Additionally, researchers might investigate whether machine learning techniques can identify non-linear relationships or optimal combinations of ELS with other signals to further improve predictive power.

In conclusion, this paper establishes Earnings Leverage Stability as a robust and

economically significant predictor of stock returns that adds value beyond existing factor models. While questions remain about its theoretical underpinnings and broader applicability, the evidence strongly supports its inclusion in the set of reliable cross-sectional return predictors.

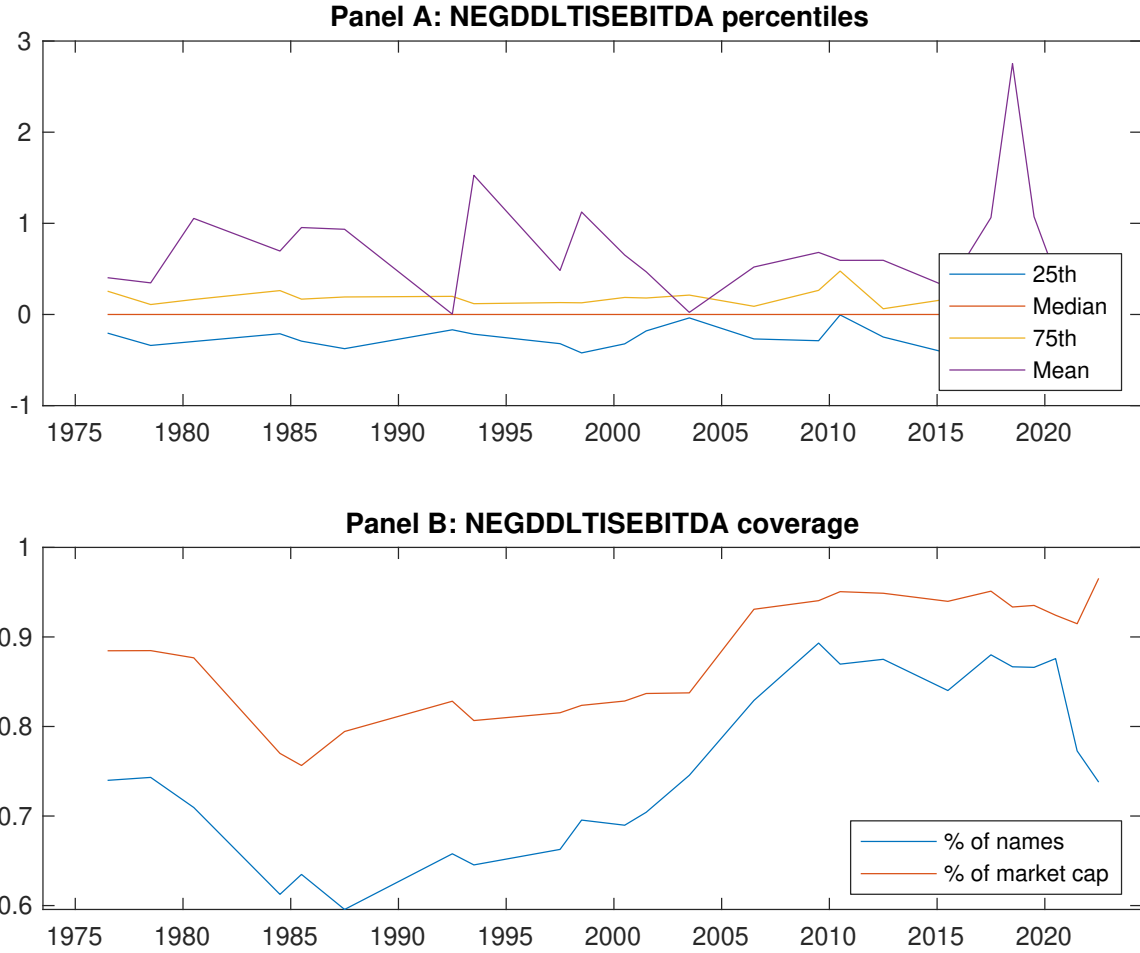


Figure 1: Times series of ELS percentiles and coverage.
This figure plots descriptive statistics for ELS. Panel A shows cross-sectional percentiles of ELS over the sample. Panel B plots the monthly coverage of ELS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ELS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on ELS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.53 [2.43]	0.64 [3.55]	0.68 [3.46]	0.74 [4.24]	0.77 [3.73]	0.25 [3.33]
α_{CAPM}	-0.21 [-3.52]	0.03 [0.61]	0.02 [0.29]	0.15 [3.32]	0.07 [1.26]	0.28 [3.83]
α_{FF3}	-0.24 [-4.21]	-0.01 [-0.31]	0.07 [1.41]	0.14 [3.25]	0.06 [1.03]	0.30 [4.07]
α_{FF4}	-0.21 [-3.61]	-0.00 [-0.09]	0.11 [2.10]	0.11 [2.49]	0.06 [1.12]	0.27 [3.66]
α_{FF5}	-0.22 [-3.80]	-0.06 [-1.49]	0.10 [1.77]	0.06 [1.35]	0.07 [1.32]	0.30 [3.95]
α_{FF6}	-0.20 [-3.39]	-0.05 [-1.24]	0.12 [2.28]	0.04 [0.91]	0.08 [1.40]	0.28 [3.68]
Panel B: Fama and French (2018) 6-factor model loadings for ELS-sorted portfolios						
β_{MKT}	1.10 [80.87]	0.98 [100.50]	0.98 [78.47]	0.96 [96.66]	1.04 [79.87]	-0.06 [-3.49]
β_{SMB}	0.13 [6.46]	-0.12 [-7.96]	-0.03 [-1.51]	-0.03 [-2.31]	0.18 [9.03]	0.04 [1.70]
β_{HML}	0.09 [3.43]	0.13 [7.05]	-0.15 [-6.19]	-0.04 [-1.99]	-0.02 [-0.62]	-0.11 [-3.13]
β_{RMW}	0.04 [1.32]	0.11 [5.94]	-0.01 [-0.31]	0.11 [5.50]	-0.05 [-2.02]	-0.09 [-2.54]
β_{CMA}	-0.11 [-2.76]	0.03 [1.12]	-0.05 [-1.37]	0.17 [5.75]	0.02 [0.66]	0.13 [2.64]
β_{UMD}	-0.04 [-2.67]	-0.02 [-1.61]	-0.04 [-3.51]	0.03 [3.01]	-0.01 [-0.61]	0.03 [1.63]
Panel C: Average number of firms (n) and market capitalization (me)						
n	669	527	1047	577	656	
me (\$10 ⁶)	1232	3166	2378	3251	1274	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ELS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.25 [3.33]	0.28 [3.83]	0.30 [4.07]	0.27 [3.66]	0.30 [3.95]	0.28 [3.68]
Quintile	NYSE	EW	0.19 [4.37]	0.21 [4.87]	0.21 [4.84]	0.19 [4.43]	0.20 [4.75]	0.20 [4.51]
Quintile	Name	VW	0.19 [2.61]	0.24 [3.23]	0.26 [3.56]	0.24 [3.24]	0.23 [3.13]	0.22 [2.97]
Quintile	Cap	VW	0.22 [3.47]	0.26 [4.02]	0.28 [4.37]	0.24 [3.69]	0.21 [3.31]	0.19 [2.90]
Decile	NYSE	VW	0.27 [2.83]	0.30 [3.14]	0.31 [3.27]	0.30 [3.16]	0.32 [3.30]	0.32 [3.25]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.18 [2.47]	0.22 [2.99]	0.23 [3.17]	0.22 [2.97]	0.23 [3.07]	0.21 [2.89]
Quintile	NYSE	EW	-0.07 [-1.26]					
Quintile	Name	VW	0.13 [1.75]	0.17 [2.39]	0.19 [2.65]	0.18 [2.49]	0.17 [2.34]	0.16 [2.19]
Quintile	Cap	VW	0.17 [2.66]	0.21 [3.26]	0.22 [3.53]	0.20 [3.18]	0.17 [2.66]	0.15 [2.38]
Decile	NYSE	VW	0.19 [1.99]	0.21 [2.20]	0.22 [2.32]	0.22 [2.29]	0.22 [2.27]	0.21 [2.22]

Table 3: Conditional sort on size and ELS

This table presents results for conditional double sorts on size and ELS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ELS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ELS and short stocks with low ELS. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ELS Quintiles					ELS Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.66 [2.44]	0.90 [3.24]	0.93 [3.43]	0.93 [3.21]	0.74 [2.73]	0.08 [1.03]	0.09 [1.16]	0.08 [1.06]	0.04 [0.53]	0.05 [0.69]	0.03 [0.35]
	(2)	0.71 [2.69]	0.84 [3.39]	0.88 [3.47]	0.89 [3.62]	0.87 [3.38]	0.16 [2.22]	0.19 [2.54]	0.16 [2.16]	0.17 [2.25]	0.17 [2.23]	0.18 [2.31]
	(3)	0.77 [3.10]	0.84 [3.81]	0.79 [3.30]	0.84 [3.78]	0.87 [3.64]	0.10 [1.22]	0.13 [1.69]	0.13 [1.66]	0.11 [1.41]	0.15 [1.82]	0.13 [1.62]
	(4)	0.64 [2.78]	0.84 [4.05]	0.85 [3.86]	0.75 [3.61]	0.91 [4.16]	0.27 [3.67]	0.30 [4.04]	0.32 [4.27]	0.30 [3.93]	0.34 [4.38]	0.32 [4.13]
	(5)	0.47 [2.32]	0.59 [3.36]	0.64 [3.27]	0.69 [3.85]	0.72 [3.71]	0.25 [2.86]	0.28 [3.14]	0.31 [3.47]	0.28 [3.13]	0.24 [2.69]	0.23 [2.54]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ELS Quintiles					ELS Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	390	391	392	391	388	36	34	33	34	35	
	(2)	106	107	106	107	106	60	61	60	61	61	
	(3)	76	76	76	76	76	107	109	104	107	106	
	(4)	64	64	64	64	63	226	240	228	236	226	
(5)	58	59	59	59	59	1322	2164	1961	2271	1420		

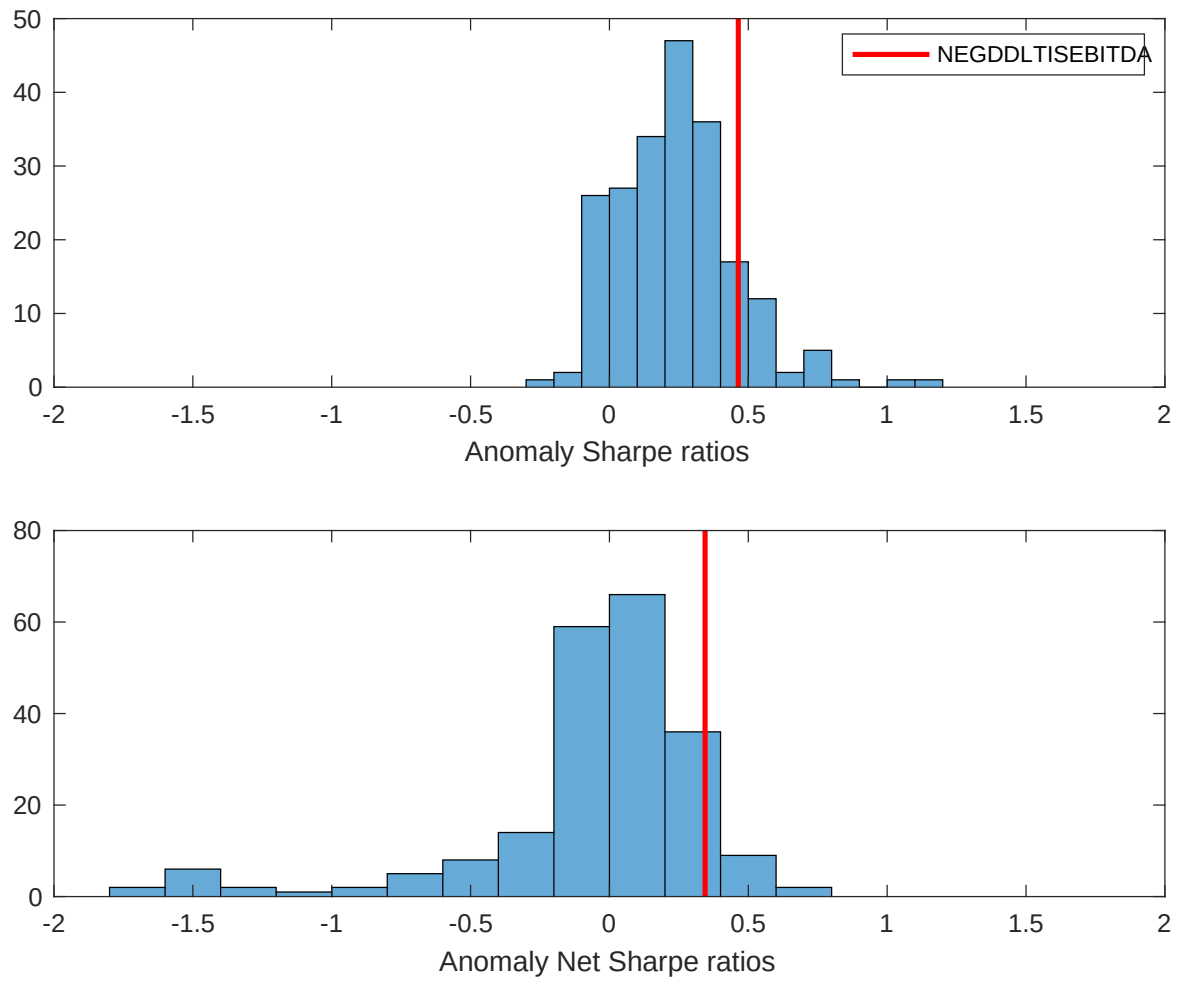


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ELS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

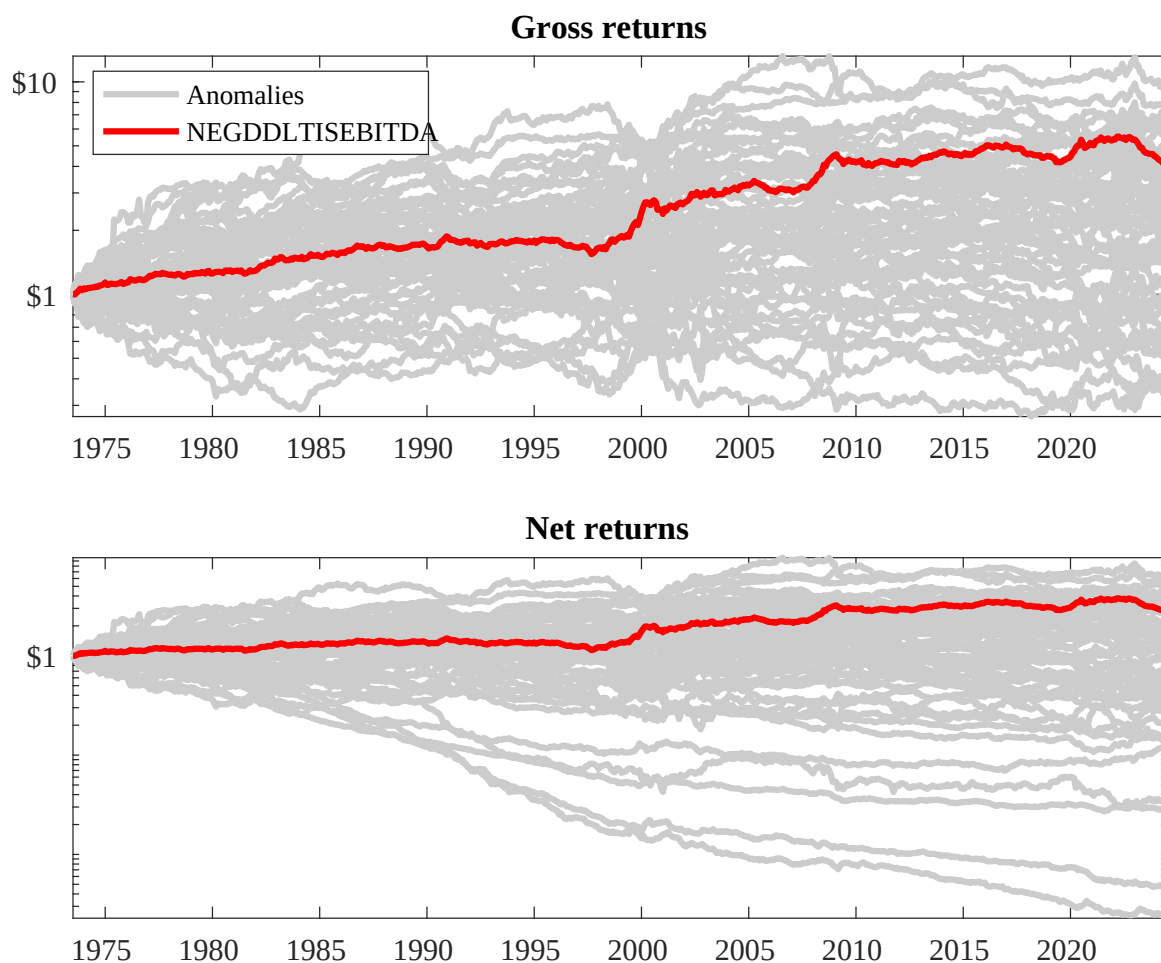
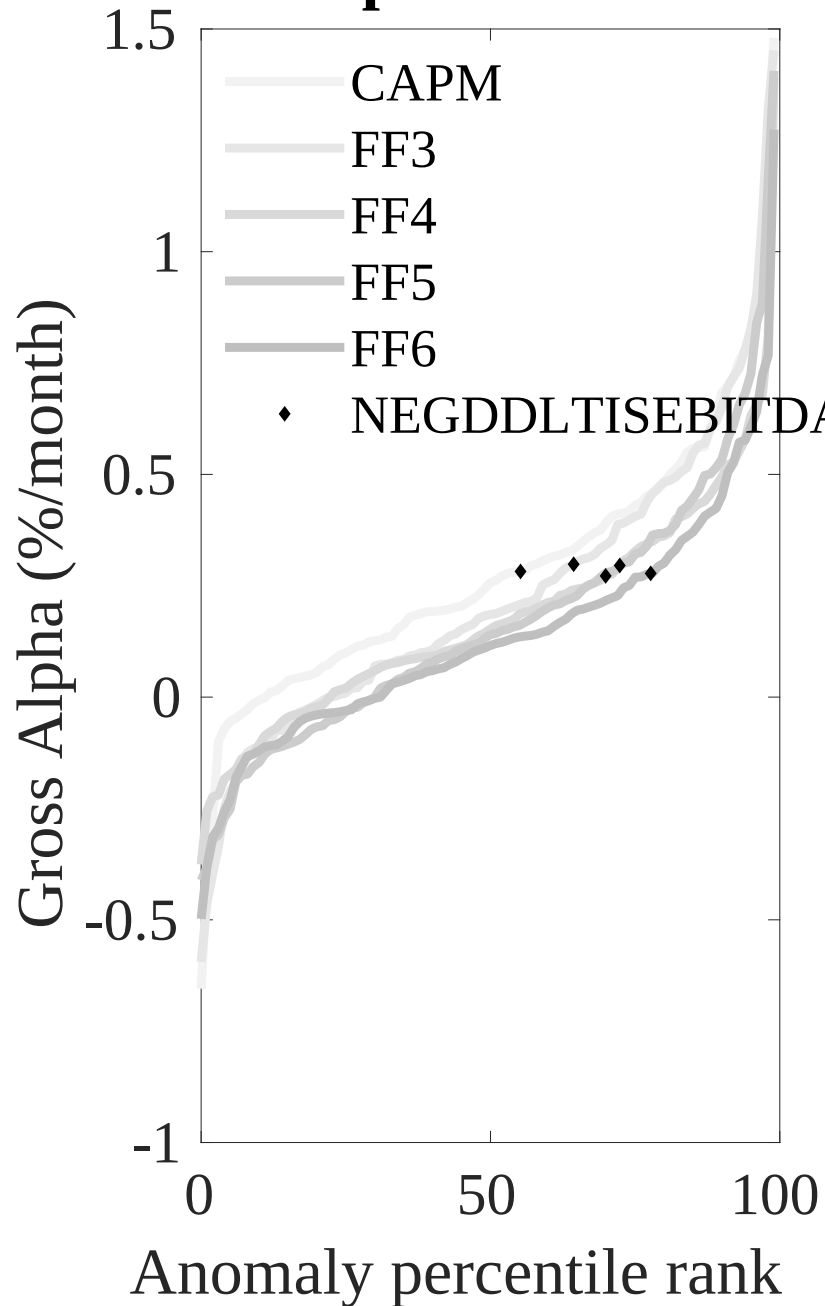


Figure 3: Dollar invested.

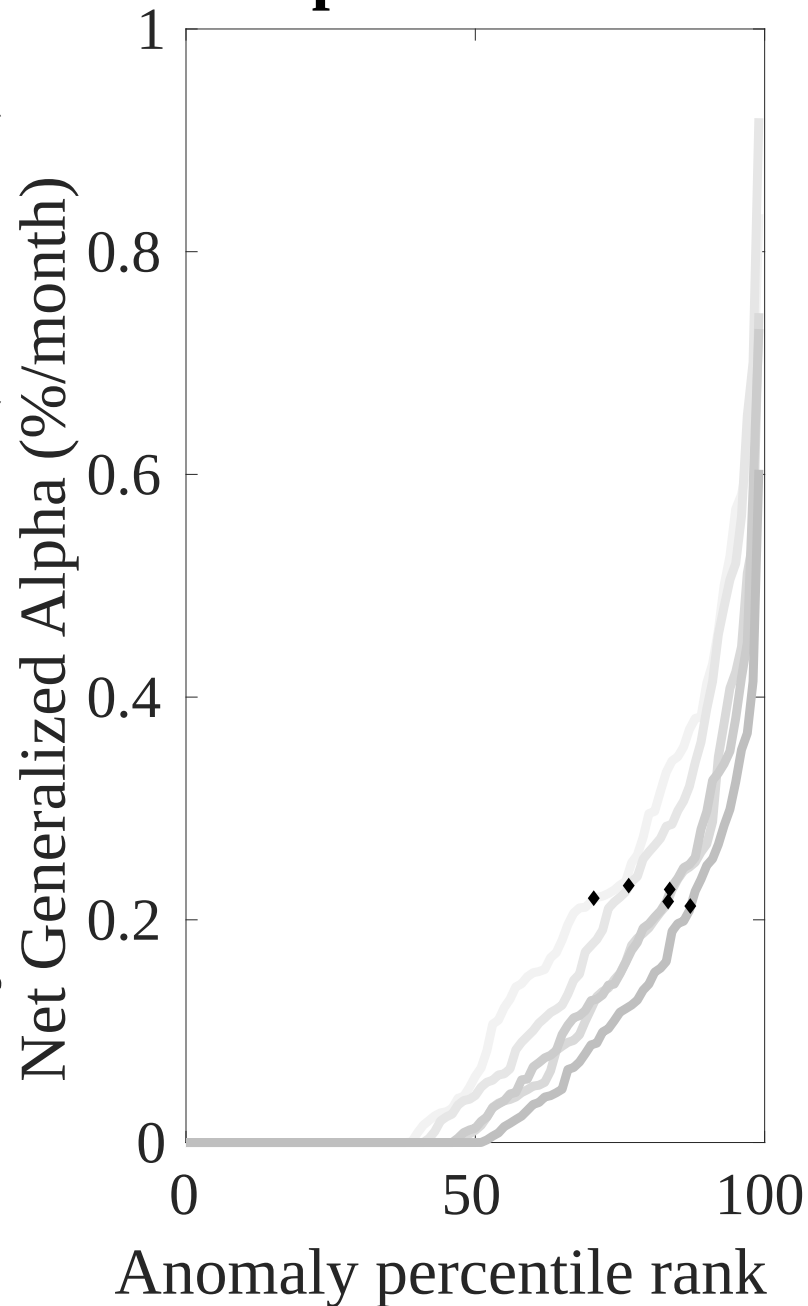
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ELS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



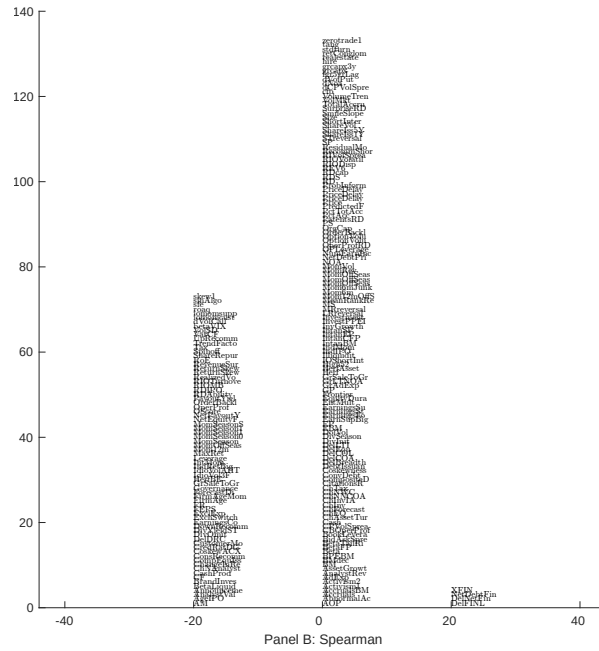
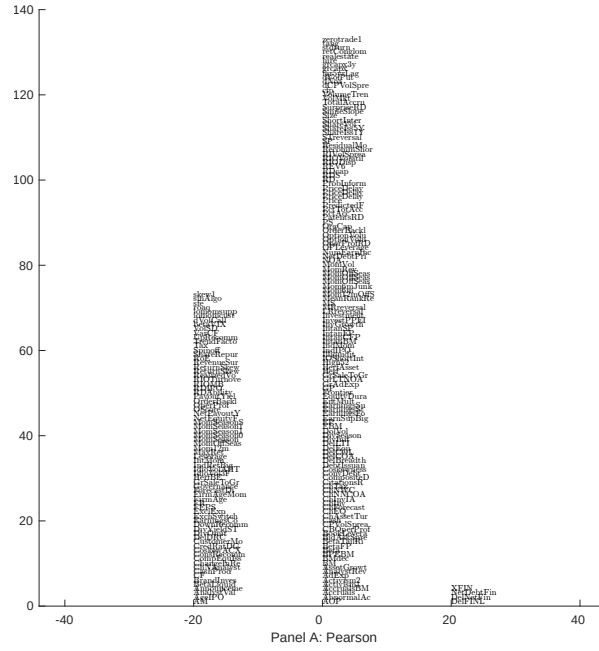


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with ELS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

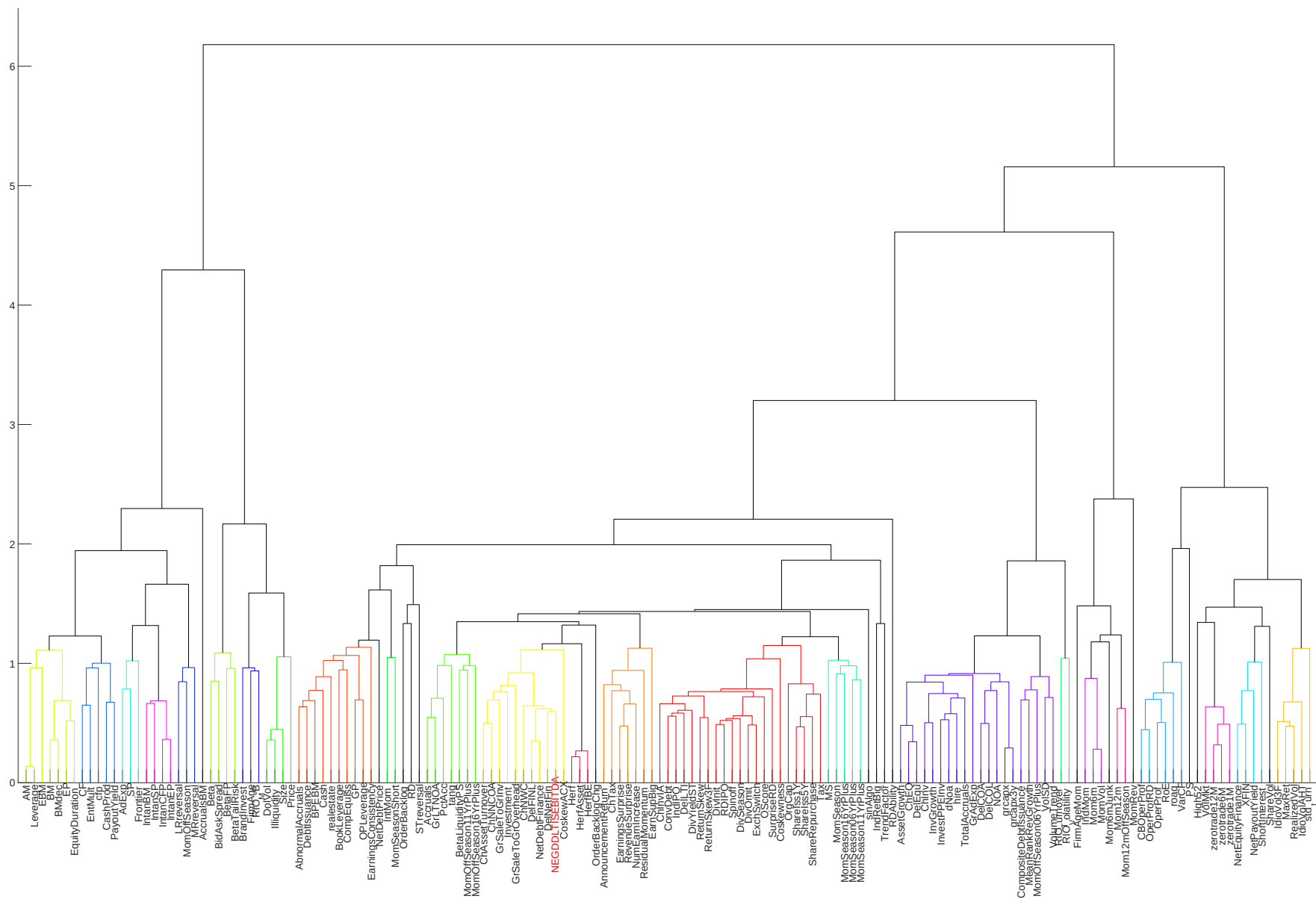


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

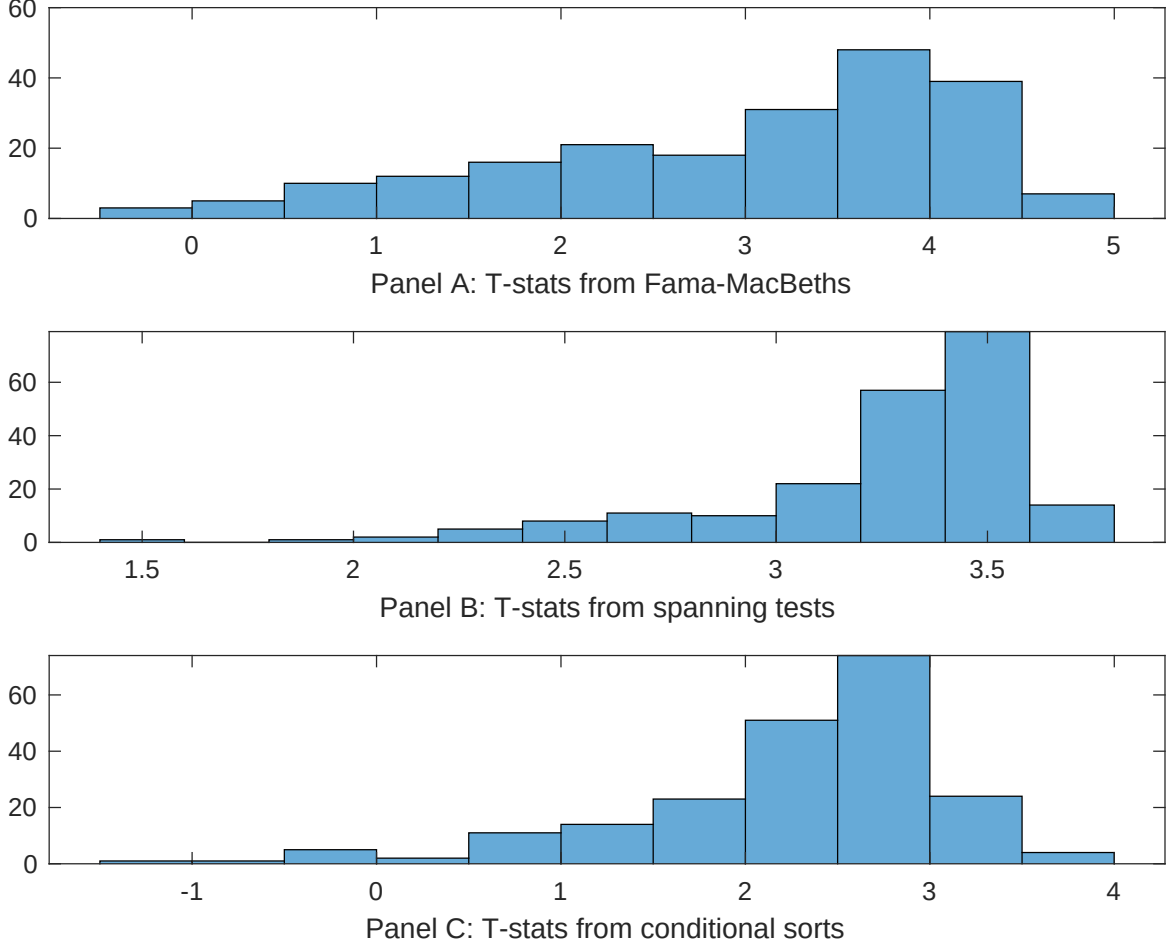


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ELS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELS} ELS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ELS. Stocks are finally grouped into five ELS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ELS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ELS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ELS} ELS_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Change in net financial assets, Accruals, Inventory Growth, Composite debt issuance. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.32]	0.13 [5.36]	0.13 [5.22]	0.12 [4.99]	0.13 [5.29]	0.14 [5.96]	0.13 [5.64]
ELS	0.91 [1.21]	0.79 [1.08]	0.18 [2.39]	0.24 [3.27]	0.30 [3.57]	0.17 [1.89]	0.72 [0.67]
Anomaly 1	0.20 [9.24]						0.21 [3.94]
Anomaly 2		0.18 [9.52]					-0.13 [-0.22]
Anomaly 3			0.73 [4.92]				-0.68 [-1.76]
Anomaly 4				0.14 [4.41]			0.91 [1.94]
Anomaly 5					0.42 [7.24]		0.23 [2.91]
Anomaly 6						0.11 [5.84]	0.53 [2.86]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ELS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ELS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Change in net financial assets, Accruals, Inventory Growth, Composite debt issuance. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.25 [3.50]	0.25 [3.41]	0.23 [3.17]	0.24 [3.25]	0.28 [3.72]	0.28 [3.76]	0.22 [3.06]
Anomaly 1	24.19 [6.09]						17.44 [3.37]
Anomaly 2		21.67 [5.22]					-1.95 [-0.33]
Anomaly 3			20.17 [5.84]				12.64 [3.27]
Anomaly 4				10.64 [3.62]			7.43 [2.56]
Anomaly 5					5.85 [2.02]		5.01 [1.80]
Anomaly 6						10.70 [3.12]	6.46 [1.86]
mkt	-5.79 [-3.47]	-5.34 [-3.17]	-5.46 [-3.26]	-5.03 [-2.93]	-5.96 [-3.47]	-5.27 [-3.07]	-4.75 [-2.86]
smb	3.34 [1.31]	3.26 [1.26]	6.65 [2.62]	7.50 [2.85]	6.13 [2.34]	3.66 [1.38]	5.72 [2.14]
hml	-8.65 [-2.68]	-8.19 [-2.51]	-9.98 [-3.09]	-6.93 [-2.06]	-9.80 [-2.96]	-11.06 [-3.33]	-8.49 [-2.58]
rmw	-11.12 [-3.37]	-10.42 [-3.14]	-6.05 [-1.83]	-4.97 [-1.43]	-7.62 [-2.25]	-9.49 [-2.82]	-6.14 [-1.77]
cma	6.89 [1.38]	5.92 [1.15]	18.32 [3.67]	9.74 [1.92]	7.95 [1.41]	10.26 [2.02]	4.01 [0.68]
umd	1.45 [0.86]	1.13 [0.66]	2.66 [1.60]	2.20 [1.29]	2.41 [1.39]	2.57 [1.51]	0.47 [0.28]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	12	10	11	8	7	8	15

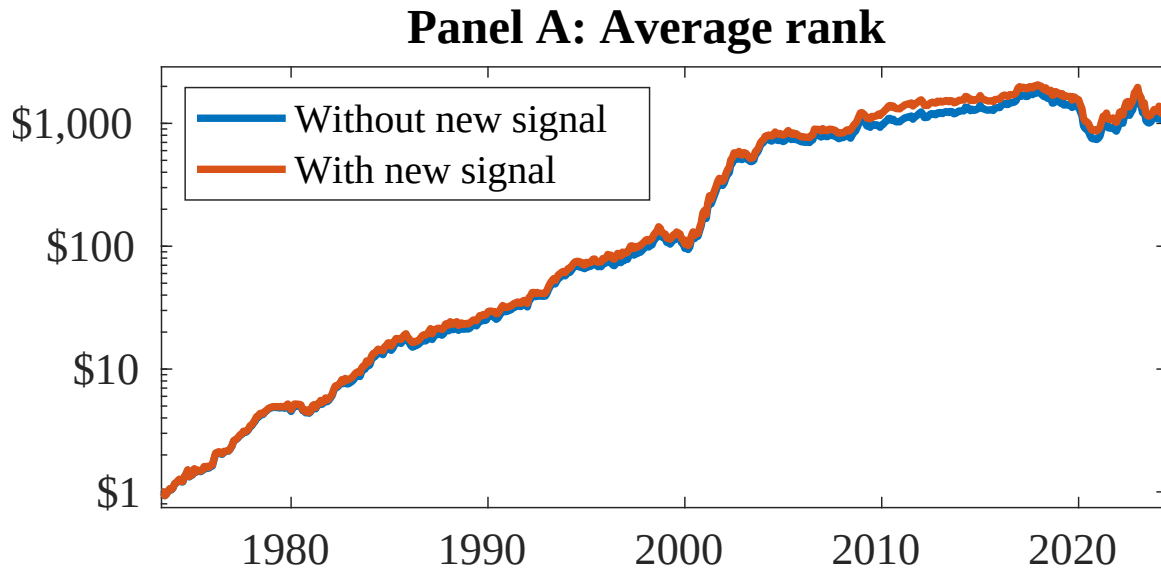


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as ELS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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